THE SECOND COEFFICIENT OF H^p FUNCTIONS WITH PRESCRIBED VALUE AT THE ORIGIN

ØISTEIN SØVIK

ABSTRACT. For $0 < p < 1$ and $0 \le t \le 1$, let $\Phi_2(p,t)$ denote the supremum of $\operatorname{Re} f''(0)/2$ over functions f in the unit ball of the Hardy space H^p with $f(0) = t$. We determine $\Phi_2(p,t)$ and all of its extremals for every p and t. In particular, we confirm the threshold suggested by Brevig, Grepstad and Instanes above which $\Phi_2(p,t) = \Phi_1(p,t)$, and we identify the point σ_p below which the extremals have a zero in the unit disc.

1. INTRODUCTION

Let H^p denote the classical Hardy space of analytic functions in the unit disc $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$, which for $0 < p < \infty$ is equipped with

$$\|f\|_{H^p}^p = \lim_{r \to 1^-} \int_0^{2\pi} |f(re^{i\theta})|^p \, \frac{\mathrm{d}\theta}{2\pi}.$$

Suppose that $k \ge 1$ is an integer and that $0 \le t \le 1$. Brevig, Grepstad and Instanes [2] studied the extremal problem

$$\Phi_k(p,t) = \sup\Big\{ \operatorname{Re} \frac{f^{(k)}(0)}{k!} : \|f\|_{H^p} \le 1 \text{ and } f(0) = t \Big\}.$$

By a standard normal families argument, there are functions attaining the supremum, which we call extremals, and every extremal satisfies $\|f\|_{H^p} = 1$ [2, Lemma 3]. For $0 < p < 1$ and $k = 1$, the problem was studied by Kabaila [4,5] and Connelly [3, Section 4], and solved completely in [2, Theorem 12]. There is $0 < t_p < 1/2$ such that the extremal has one zero in $\mathbb{D}$ when $t < t_p$ and none when $t > t_p$, and Kabaila showed that t_p is the only value of t for which there are two extremals, one with a zero and one without. For $k = 2$, Brevig, Grepstad and Instanes treated the zero-free extremals analytically and the extremals with zeros numerically, and their computations suggested a threshold above which $\Phi_2(p,t) = \Phi_1(p,t)$ [2, Section 6]. In [6] we proved that for every k an extremal has at most one zero in $\mathbb{D}$. The main purpose of the present paper is to solve the case $k = 2$ completely.

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Throughout the paper we assume that $0 < p < 1$ and set $q = 2/p - 1 > 1$ and $\nu = 2/p$, and for $0 < t \le 1$ we write $t = (1+\beta^2)^{-1/p}$ with $\beta \ge 0$. Our first result describes the extremals.

Theorem 1. *There are numbers $0 < \sigma_p < t_\star < 1$ such that the extremals for $\Phi_2(p,t)$, $0 < t < 1$, are as follows:*

(i) *if $t < \sigma_p$, there are exactly two extremals, and each has one zero in $\mathbb{D}$;*
(ii) *if $t = \sigma_p$, there are exactly four extremals: two with one zero and two without zeros;*
(iii) *if $\sigma_p < t < t_\star$, there are exactly two extremals, and they have no zeros;*
(iv) *if $t \ge t_\star$, there is exactly one extremal, and it has no zeros.*

Moreover, $\Phi_2(p,t) = \Phi_1(p,t)$ if and only if $t_\star \le t \le 1$, and the right derivative of $\Phi_2(p,t)$ at σ_p is strictly larger than the left derivative.

The two transitions are of different kinds. At σ_p the one-zero extremals still have their zero strictly inside $\mathbb{D}$; they do not become zero-free by moving the zero to the boundary. Instead a different family of extremals overtakes them, which is why four extremals coexist and the value has a corner. At $t_\star$ the two zero-free extremals $t(1 \pm xz + q^{-1}z^2)^\nu$ merge, because their linear coefficient x tends to 0.

Our second result gives the extremals and the value exactly in every regime. Put

$$t_\star = \Big(1 + \big(\frac{p}{2-p}\big)^2\Big)^{-1/p},$$

so that $\beta = 1/q$ at $t = t_\star$. For $1/q \le A < q^{-1/2}$ let

$$r = \frac{1 - qA^2}{(q-1)A}, \qquad N = 1 + (A+r)^2 + A^2 r^2, \qquad \tau(A) = rN^{-1/p},$$

and for $1/q < A < q^{-1/2}$ let

$$f_A(z) = \frac{(z+r)(1+rz)^{2/p-1}(1+Az)^{2/p}}{N^{1/p}},$$

so that $0 < r < 1$, $f_A(0) = \tau(A)$, and f_A has a single zero $-r$ in $\mathbb{D}$.

Theorem 2. *The function*

$$D(A) = q + \frac{\nu A}{r} + q\nu A r + \binom{\nu}{2} A^2 + \binom{q}{2} r^2 - \frac{1 + q^2(\tau(A)^{-p} - 1)}{pq}$$

has exactly one zero A_c in $(1/q, q^{-1/2})$, and $\sigma_p = \tau(A_c)$. Moreover:

(a) *if $0 < t \le \sigma_p$ and A is the largest solution of $\tau(A) = t$ in $(1/q, q^{-1/2})$, then*

$$\Phi_2(p,t) = \frac{f_A''(0)}{2} = t\Big(q + \frac{\nu A}{r} + q\nu A r + \binom{\nu}{2} A^2 + \binom{q}{2} r^2\Big),$$

and the one-zero extremals are $f_A(z)$ and $f_A(-z)$;

(b) *if $\sigma_p \le t \le t_\star$, then*

$$\Phi_2(p,t) = \frac{t}{pq}(1+q^2\beta^2),$$

and the zero-free extremals are $t(1\pm xz+q^{-1}z^2)^\nu$ with $x=\sqrt{\beta^2-q^{-2}}$;

(c) *if $t_\star \le t \le 1$, then $\Phi_2(p,t)=\Phi_1(p,t)=t\nu\beta$, and the extremal is $t(1+\beta z^2)^\nu$.*

For $t=0$ we have $\Phi_2(p,0)=C(1,p)=\sqrt{2/p}\,(1-p/2)^{1/p-1/2}$, the norm of $f\mapsto f'(0)$ on H^p computed in [1]*, and the extremals are zF with F an extremal for $C(1,p)$.*

In (a), A is the inverse of the explicit function τ on the branch where it decreases from $\tau(A_c)=\sigma_p$ to 0. The number $tD(A)$ is the difference between the values in (a) and (b), so σ_p is the point where the two kinds of candidates give the same value. It plays the role of t_p for $k=1$. For instance, $\sigma_{2/3}=0.1562\ldots$ and $t_\star=0.7155\ldots$ when $p=2/3$. We record two consequences of the formulas which can be read without them.

Corollary 1. *As $t\to 0^+$,*

$$\Phi_2(p,t)=C(1,p)+\Big(\frac{1}{p}-1\Big)t+O(t^2).$$

Corollary 2. *Let $\psi(r)=r\big(a(1+r^2)+br\big)^{-1/p}$ with $a=1+q^{-2}$ and $b=2/q$, and let $r_*\in(0,1)$ be the positive root of $(2-p)ar^2+(1-p)br-pa=0$. Then*

$$\big(2+2q^{-1}+2q^{-2}\big)^{-1/p}=\psi(1)<\sigma_p<\psi(r_*)<t_\star.$$

In particular $\sigma_p\to 1/6$ as $p\to 1^-$, while $t_\star\to 1/2$.

Let us briefly describe our approach, which has four steps. By [6], an extremal is gh^q with g and h quadratic polynomials, and optimizing over g leaves an explicit function of the two coefficients of h (Section 2). A zero-free extremal is then h^ν, and the problem becomes the maximization of a concave quadratic in the size of the coefficient of z^2; this produces the transition at $t_\star$. A one-zero extremal is pinned down by the first-order condition to two one-parameter families, one with real and one with purely imaginary coefficients. Adding a small real part to the middle coefficient improves the imaginary family, which leaves $f_A(\pm z)$ (Section 3). Finally, the two kinds of candidates are compared through D. The key identity is that D' and the derivative of τ^{-p} are proportional with a positive factor, so D and τ move in opposite directions. This locates the unique crossing σ_p (Section 4).

Organization. The present paper is organized into five additional sections. In Section 2 we reduce the problem to an explicit function of two complex variables, and in Section 3 we determine the candidates for the extremals. Section 4 compares the two kinds of candidates, and the main results are

proved in Section 5. Section 6 contains some concluding remarks and open problems.

2. Reduction to quadratic factors

Let f be an extremal for $\Phi_2(p,t)$ with $0<t<1$. As in [6], write $f=IO$ with I inner and O outer, and put $h=O^{p/2}$ and $g=Ih$, normalized so that $h(0)>0$. Then $f=gh^q$, $|g|=|h|$ on $\mathbb{T}$, and $\|g\|_{H^2}=\|h\|_{H^2}=1$. If g were constant, then $g=1$ and $h=1$, so $f=1$, which contradicts $f(0)<1$. Hence, by [6, Lemma 1], g is a polynomial of degree 2, and since $|h|=|g|$ on $\mathbb{T}$, so is h. Moreover there is $\mu>0$ such that

$$\mu g^*(z)=h(z)^q+O(z^2),\qquad g^*(z)=z^2\overline{g(1/\overline{z})}, \tag{1}$$

and by [6, Theorem 1], f has at most one zero in $\mathbb{D}$.

For a fixed h, the best first factor is found by the Cauchy–Schwarz inequality [6, Section 2]. For quadratic h this gives an explicit function of its coefficients. Write

$$h(z)=\frac{1+bz+cz^2}{\sqrt{K}},\qquad K=1+|b|^2+|c|^2.$$

The first three coefficients of h^q are

$$K^{-q/2},\qquad qbK^{-q/2},\qquad \Big(qc+\frac{q(q-1)}{2}b^2\Big)K^{-q/2},$$

so the reduced objective $\Lambda(h)$ of [6, Section 2] is

$$\Lambda_t(b,c)=t\,\mathrm{Re}\Big(qc+\frac{q(q-1)}{2}b^2\Big)+\sqrt{K^{-q}-t^2}\,\sqrt{1+q^2|b|^2}, \tag{2}$$

defined when h has no zeros in $\mathbb{D}$ and $K^{-q}>t^2$. For every such (b,c) we have $\Lambda_t(b,c)\le\Phi_2(p,t)$, and if $f=gh^q$ is an extremal, then $\Lambda_t(b,c)=\Phi_2(p,t)$ for the coefficients of its factor h [6, Lemma 1].

3. The candidates

3.1. **Zero-free extremals.** Suppose that f has no zeros in $\mathbb{D}$. Then the polynomial g has no zeros in $\mathbb{D}$ and is therefore outer, so $g=h$, since $|g|=|h|$ on $\mathbb{T}$ and $g(0)$, $h(0)>0$. Hence $f=h^\nu$ with $h(0)=t^{p/2}$. Write $h(z)=t^{p/2}(1+u_1z+u_2z^2)$. The condition $\|h\|_{H^2}=1$ gives $|u_1|^2+|u_2|^2=\beta^2$, and the coefficient of z^2 in f is $t\nu(u_2+qu_1^2/2)$. With $x=|u_2|$, the size of the quadratic coefficient,

$$\frac{\mathrm{Re}\,f''(0)}{2t}\le\nu\Big(x+\frac{q}{2}(\beta^2-x^2)\Big),\qquad 0\le x\le\beta,$$

with equality if and only if $u_2\ge0$ and u_1 is real. The right-hand side is a concave quadratic in x with its maximum at $x=\min(\beta,1/q)$. If $\beta\le1/q$, all the available coefficient norm goes into u_2: then $\mathrm{Re}\,f''(0)/2\le t\nu\beta$, with

equality only for $f(z) = t(1+\beta z^2)^\nu$. If $\beta > 1/q$, some of it remains for the linear term, with two possible signs:

$$\frac{\operatorname{Re} f''(0)}{2} \le \frac{t}{pq}(1+q^2\beta^2), \tag{3}$$

with equality only for the two functions $f_\pm(z) = t(1 \pm xz + q^{-1}z^2)^\nu$, where $x = \sqrt{\beta^2 - q^{-2}}$.

These functions are admissible only if $1 \pm xz + q^{-1}z^2$ has no zeros in $\mathbb{D}$, which holds when $x \le 1 + 1/q$, that is, when $t \ge \tau(1/q) = (2 + 2q^{-1} + 2q^{-2})^{-1/p}$. Indeed, write $1 + xz + q^{-1}z^2 = (1+\alpha z)(1+\gamma z)$. Either α and γ are conjugate with modulus $q^{-1/2}$, or they are positive with $\alpha\gamma = 1/q$ and $(1-\alpha)(1-\gamma) = 1 - x + 1/q \ge 0$, in which case neither exceeds 1. Then $\|f_\pm\|_{H^p}^p = t^p(1+\beta^2) = 1$.

3.2. **One-zero extremals.** Suppose that f has exactly one zero w in $\mathbb{D}$. Since $t > 0$, we have $w \ne 0$. The polynomial g has degree 2 and one zero in $\mathbb{D}$, so $g(z) = u_0(z-w)(1+az)$ with $0 < |a| \le 1$. Since h is outer with $|h| = |g|$ on $\mathbb{T}$ and $h(0) > 0$,

$$h(z) = |u_0|(1 - \overline{w}z)(1+az).$$

The first two coefficients of (1) give $\mu\overline{u_0 a} = |u_0|^q$ and $\mu\overline{u_0}(1 - \overline{a}\,\overline{w}) = q|u_0|^q(a - \overline{w})$. Dividing the second identity by the first and taking conjugates yields

$$(q-1)aw = q|a|^2 - 1. \tag{4}$$

The first identity shows that $u_0 a > 0$, while $u_0 w = -g(0) < 0$. Hence $w/a < 0$, and since aw is real by (4), either a and w are real with $aw < 0$, or a, w and u_0 are purely imaginary with $aw > 0$. Put $A = |a|$. Then $A < 1$, since $A = 1$ would give $aw = 1$ by (4). Replacing $f(z)$ by $f(-z)$ if necessary, we may assume that $a = A$ in the first case and $a = iA$ in the second.

Lemma 1. *The case where a and w are purely imaginary does not occur.*

Proof. Here $a = iA$ and $w = -ir$ with $r > 0$, and (4) reads $(q-1)Ar = qA^2 - 1$. Normalizing h as in Section 2, we get $b = i(A-r)$, $c = Ar$ and $K = 1 + (A-r)^2 + A^2r^2$, and since $u_0 = -i|u_0|$ and $|u_0| = K^{-1/2}$, we have $t = -u_0 w|u_0|^q = rK^{-(q+1)/2}$. Put $B = A - r$. The relation $(q-1)Ar = qA^2 - 1$ gives $qAB = 1 - Ar$, and hence

$$1 + q^2B^2 = \frac{K - r^2}{A^2};$$

in particular $K > r^2$. We perturb only the middle coefficient: for real ε, put $b_\varepsilon = b + \varepsilon$ and $K_\varepsilon = K + \varepsilon^2$. The zeros of $1 + bz + cz^2$ are $-i/r$ and i/A, both outside $\overline{\mathbb{D}}$, and $K^{-q} - t^2 = (K - r^2)K^{-(q+1)} > 0$, so (b_ε, c) is admissible in (2) for small ε, and $\Lambda_t(b_\varepsilon, c) \le \Phi_2(p,t) = \Lambda_t(b,c)$. We show that

$$\Lambda_t(b_\varepsilon, c) = \Lambda_t(b,c) + q(q-1)t\,\varepsilon^2 + O(\varepsilon^4), \tag{5}$$

which is a contradiction. Since $\operatorname{Re} b_\varepsilon^2 = \varepsilon^2 - B^2$ and $|b_\varepsilon|^2 = B^2 + \varepsilon^2$, the first term of (2) increases by $q(q-1)t\varepsilon^2/2$. The second term of (2) is a function of $s = \varepsilon^2$, and its logarithmic derivative at $s = 0$ is

$$\frac{1}{2}\Big(-\frac{qK^{-q-1}}{K^{-q} - t^2} + \frac{q^2}{1 + q^2B^2}\Big) = \frac{q}{2} \cdot \frac{qA^2 - 1}{K - r^2}.$$

At $s = 0$ the second term equals $(K - r^2)K^{-(q+1)/2}/A$, so its derivative is $q(qA^2-1)K^{-(q+1)/2}/(2A) = q(q-1)rK^{-(q+1)/2}/2 = q(q-1)t/2$. Adding the two contributions gives (5); the remainder is $O(\varepsilon^4)$ since everything depends on ε^2 only. □

Hence $a = A$ and w are real with $aw < 0$, so that $w = -r$ with $r > 0$ and $u_0 > 0$. By (4), $r = (1 - qA^2)/((q-1)A)$, and $0 < r < 1$ if and only if $1/q < A < q^{-1/2}$. The condition $\|h\|_{H^2} = 1$ gives $u_0 = N^{-1/2}$, and therefore $f = u_0^\nu(z + r)(1 + rz)^q(1 + Az)^\nu = f_A$ and $t = \tau(A)$. Conversely, f_A is admissible for $\Phi_2(p, \tau(A))$ for every $1/q < A < q^{-1/2}$, since $\|f_A\|_{H^p}^p = \|h\|_{H^2}^2 = 1$. Moreover $N > 1 + A^2 > 1 + q^{-2}$ and $r < 1$, so that

$$\tau(A) < (1 + q^{-2})^{-1/p} = t_\star. \tag{6}$$

Expanding $(1 + z/r)(1 + rz)^q(1 + Az)^\nu$ shows that

$$\frac{f_A''(0)}{2\tau(A)} = E(A), \qquad E(A) = q + \frac{\nu A}{r} + q\nu Ar + \binom{\nu}{2}A^2 + \binom{q}{2}r^2, \tag{7}$$

so that $D(A) = E(A) - (1 + q^2(\tau(A)^{-p} - 1))/(pq)$, and $\tau(A)D(A)$ is the value of f_A minus the right-hand side of (3) at $t = \tau(A)$. We use this formula to define D on $[1/q, q^{-1/2})$.

4. Comparison of the two kinds of candidates

In preparation for the comparison, we collect three facts about τ, D and $M(A) = 1/(Ar^{1-p})$. Everything in the comparison follows from them.

Lemma 2. (i) *There is $A_m \in (1/q, q^{-1/2})$ such that τ is strictly increasing on $[1/q, A_m]$ and strictly decreasing on $[A_m, q^{-1/2}]$, with $\tau(q^{-1/2}) = 0$.*
(ii) *$M(1/q) = q$, and M is strictly increasing on $[1/q, q^{-1/2})$.*
(iii) *On $(1/q, q^{-1/2})$,*

$$D'(A) = \frac{M(A) - q}{p} \frac{\mathrm{d}}{\mathrm{d}A}\tau(A)^{-p}. \tag{8}$$

By (ii) and (iii), D moves in the opposite direction to τ: it decreases on the branch where τ increases and increases on the branch where τ decreases. So only the decreasing branch of τ can supply a winning one-zero candidate.

Proof. Put $y = A^2$. A direct computation gives

$$\frac{\mathrm{d}}{\mathrm{d}y}\log\tau(\sqrt{y}) = \frac{Q(y)}{2(q-1)^2y^2(1 - qy)N}, \tag{9}$$

where

$$Q(y) = (2q^4 + q^3)y^4 - (4q^3 + 2q^2)y^3 + (-q^3 + 4q^2 + 3q - 2)y^2 - 2q^2y + q.$$

For $0 \le x \le 1$,

$$Q'(x/q) = -2(1-x)\Big(2(2q+1)\Big(x-\frac{1}{4}\Big)^2+\frac{14q-9}{8}\Big)-2(q-1)^2\Big(1+x+\frac{2x}{q}\Big) < 0,$$

so Q is strictly decreasing on $[0, 1/q]$. Since $Q(q^{-2}) = (q-1)^4(q+1)^2/q^5 > 0$ and $Q(q^{-1}) = -2(q-1)^2(q+1)/q^2 < 0$, the function $Q(A^2)$ changes sign exactly once on $(1/q, q^{-1/2})$, from positive to negative, and (9) gives (i); $r = 0$ at $A = q^{-1/2}$.

For (ii), $r = 1$ at $A = 1/q$, and $M'(A)/M(A) = 2(q^2A^2 - 1)/(\nu A(1 - qA^2)) > 0$.

For (iii), $(q-1)Ar = 1 - qA^2$ gives $r'(A) = -(1 + qA^2)/((q-1)A^2)$, and inserting this into the derivative of E yields

$$E'(A) = -\frac{Q(A^2)}{(q-1)A^3(1-qA^2)^2}.$$

On the other hand, (9) and $\tau^{-p} = r^{-p}N$ give

$$\frac{\mathrm{d}}{\mathrm{d}A}\tau(A)^{-p} = -p\tau(A)^{-p}\frac{\mathrm{d}}{\mathrm{d}A}\log\tau(A) = -\frac{pQ(A^2)}{(q-1)^2A^3(1-qA^2)r^p}.$$

Dividing the two identities and using $1 - qA^2 = (q-1)Ar$ once more, we find $E' = (M/p)(\tau^{-p})'$. Since $D' = E' - (q/p)(\tau^{-p})'$, this is (8). □

Lemma 3. *D has exactly one zero A_c in $(1/q, q^{-1/2})$. It satisfies $A_c > A_m$, $D < 0$ on $(1/q, A_c]$ and $D > 0$ on $(A_c, q^{-1/2})$, and*

$$\tau(1/q) < \tau(A_c) < \tau(A_m) < t_\star. \tag{10}$$

Proof. Using $r = 1$ and $\tau(1/q)^{-p} = 2 + 2q^{-1} + 2q^{-2}$ at $A = 1/q$, one checks that $D(1/q) = 0$. By Lemma 2, D is strictly decreasing on $[1/q, A_m]$ and strictly increasing on $[A_m, q^{-1/2})$, so $D < 0$ on $(1/q, A_m]$. Let $A_1 \in (A_m, q^{-1/2})$ be the point with $\tau(A_1) = \tau(1/q)$. Integrating (8) by parts and using $M(1/q) = q$, we obtain

$$D(A_1) = \frac{1}{p}\int_{1/q}^{A_1} M'(A)(\tau(1/q)^{-p} - \tau(A)^{-p})\,\mathrm{d}A > 0,$$

since $\tau(A) > \tau(1/q)$ for $1/q < A < A_1$. Hence D has exactly one zero A_c, and $A_m < A_c < A_1$. In particular $\tau(A_c) > \tau(A_1) = \tau(1/q)$: the crossing happens where the zero-free candidates $f_\pm$ are admissible. The remaining inequalities in (10) follow from (i) and (6). □

5. Proofs of the main results

Put $\sigma_p = \tau(A_c)$. By (10), $0 < \sigma_p < t_\star$.

Proof of Theorem 2. Let $0 < t < 1$. If $t \geq t_\star$, then f has no zeros by (6), and since $\beta \leq 1/q$, the zero-free case gives $f(z) = t(1+\beta z^2)^\nu$ and $\Phi_2(p,t) = t\nu\beta$. The function $t(1+\beta z)^\nu$ has H^p norm $t(1+\beta^2)^{1/p} = 1$, so $\Phi_1(p,t) \geq t\nu\beta$. Conversely, if f is admissible for $\Phi_1(p,t)$, then so is $f(z^2)$ for $\Phi_2(p,t)$, with coefficient of z^2 equal to $f'(0)$, so $\Phi_1(p,t) \leq \Phi_2(p,t)$. This proves (c); for $t = 1$ the only admissible function is 1.

Let $0 < t < t_\star$, so that $\beta > 1/q$. Every zero-free candidate satisfies (3). Every one-zero candidate is $f_A(\pm z)$ with $\tau(A) = t$, and its value is the right-hand side of (3) plus $tD(A)$. By Lemmas 2 and 3, if $A \leq A_m$ then $D(A) < 0$, and if $A > A_m$ then $D(A) < 0$, $D(A) = 0$ or $D(A) > 0$ according as $t > \sigma_p$, $t = \sigma_p$ or $t < \sigma_p$, since τ is decreasing there. If $\sigma_p \leq t < t_\star$, then $t > \tau(1/q)$, so $f_\pm$ are admissible and attain (3). This proves (b), with $f_{A_c}(\pm z)$ as the only other extremals when $t = \sigma_p$. If $0 < t < \sigma_p$, then $t < \tau(A_m)$, so the equation $\tau(A) = t$ has a largest solution $A > A_m$, and $D(A) > 0$. Since an extremal exists [2, Lemma 3], the extremals are $f_A(z)$ and $f_A(-z)$. This proves (a).

For $t = 0$, every admissible f is zF with $\|F\|_{H^p} = \|f\|_{H^p}$ and $f''(0)/2 = F'(0)$, so $\Phi_2(p,0) = C(1,p)$ and the extremals are zF with F an extremal for $C(1,p)$. □

Proof of Theorem 1. The list of extremals follows from Theorem 2: each f_A has the zero $-r$ in $\mathbb{D}$, and $f_\pm$ and $t(1+\beta z^2)^\nu$ have no zeros in $\mathbb{D}$.

Near σ_p, the difference between $\Phi_2(p,t)$ and the right-hand side of (3) is $tD(A(t))$ for $t < \sigma_p$ and 0 for $t \geq \sigma_p$, where $A(t)$ is the largest solution of $\tau(A) = t$. Since $\tau'(A_c) < 0$ and $D(A_c) = 0$, (8) shows that the left derivative of $tD(A(t))$ at σ_p equals $-(M(A_c) - q)\sigma_p^{-p} < 0$. Hence the right derivative of $\Phi_2(p,t)$ at σ_p exceeds the left derivative by $(M(A_c) - q)\sigma_p^{-p}$.

It remains to show that $\Phi_2(p,t) > \Phi_1(p,t)$ for $0 \leq t < t_\star$. For $t = 0$, every admissible f for $\Phi_1(p,0)$ is zF with $|f'(0)| = |F(0)| \leq \|F\|_{H^p}$, so $\Phi_1(p,0) = 1$, attained by z. The function z^2 is admissible for $\Phi_2(p,0)$ with value 1, and it is not an extremal, since by [6, Theorem 1] an extremal has only one zero in $\mathbb{D}$. For $0 < t < t_\star$, let f_1 be an extremal for $\Phi_1(p,t)$. Then $f_1(z^2)$ is admissible for $\Phi_2(p,t)$, with coefficient of z^2 equal to $\Phi_1(p,t)$, but the extremals for $\Phi_2(p,t)$ found above have a nonzero coefficient of z. Hence $f_1(z^2)$ is not an extremal, and $\Phi_2(p,t) > \Phi_1(p,t)$. □

Proof of Corollary 1. For small $t > 0$ we are in case (a) with A close to $q^{-1/2}$, that is, r close to 0. We use r as the parameter: the relation $(q-1)Ar = 1 - qA^2$ defines A as a smooth function of r near 0, with $A(0) = q^{-1/2}$ and $A'(0) = -(q-1)/(2q)$. Then $N(0) = 1 + q^{-1}$ and $N'(0) = 2A(0)(A'(0)+1) =$

$(q+1)q^{-3/2}$. Since $t = rN^{-1/p}$, we have $r = N(0)^{1/p}t + O(t^2)$. By (7),

$$\Phi_2(p,t) = \nu A N^{-1/p} + t\Big(q + q\nu A r + \binom{\nu}{2}A^2 + \binom{q}{2}r^2\Big).$$

The first term equals $C(1,p)$ at $r=0$, and its logarithmic derivative there is $A'(0)/A(0) - N'(0)/(pN(0)) = -q^{1/2}$, since $1/p = (q+1)/2$. Hence the first term is $C(1,p) - C(1,p)q^{1/2}N(0)^{1/p}t + O(t^2) = C(1,p) - \nu t + O(t^2)$. The second term is $t(q + \binom{\nu}{2}q^{-1}) + O(t^2) = t(q + \nu/2) + O(t^2)$, since $\nu - 1 = q$. As $q - \nu/2 = 1/p - 1$, this proves the expansion. □

Proof of Corollary 2. For $1/q < A < q^{-1/2}$ we have $N \geq 1 + (q^{-1} + r)^2 + q^{-2}r^2 = a(1+r^2) + br$, since the expression defining N increases with A for fixed r, so $\tau(A) \leq \psi(r)$. At $A = 1/q$ we have $r = 1$ and $\tau(1/q) = \psi(1)$. Hence, by (10), $\psi(1) < \sigma_p < \tau(A_m) \leq \sup_{0<r<1}\psi(r)$. The logarithmic derivative of ψ vanishes exactly when $(2-p)ar^2 + (1-p)br - pa = 0$, and this quadratic has one positive root r_*, which lies in $(0,1)$ since the left-hand side is $-pa < 0$ at $r = 0$ and $2(1-p)(a + b/2) > 0$ at $r = 1$. So the supremum is $\psi(r_*)$. Finally $a(1+r^2) + br > a$, so $\psi(r) < ra^{-1/p} < a^{-1/p} = t_\star$ for $0 < r < 1$.

As $p \to 1^-$ we have $a \to 2$, $b \to 2$ and $1/p \to 1$, so $\psi(1) \to 1/6$, and ψ converges uniformly on $[0,1]$ to $r/(2 + 2r + 2r^2)$, whose maximum is attained at $r = 1$ and equals $1/6$. Hence $\sigma_p \to 1/6$. Finally $t_\star \to 1/2$. □

6. Concluding remarks and further work

Numerically, σ_p is not monotone in p: it increases to about 0.168 near $p = 0.9$ and then decreases to $1/6$. It is also strictly smaller than the point t_p of [2, Theorem 12] for every p we tried, by a factor between about 0.28 and 0.44 (see Figure 1). In other words, the extremals for Φ_2 appear to stop vanishing at a smaller value of t than those for Φ_1. These observations are not proved here, and we state the second one as an open problem.

Problem 1. *Is it true that $\sigma_p < t_p$ for every $0 < p < 1$?*

The proved bounds of Corollary 2 are far apart as $p \to 0^+$, where the numerics suggest that $\sigma_p \to 0$. It would also be interesting to find a closed form for σ_p.

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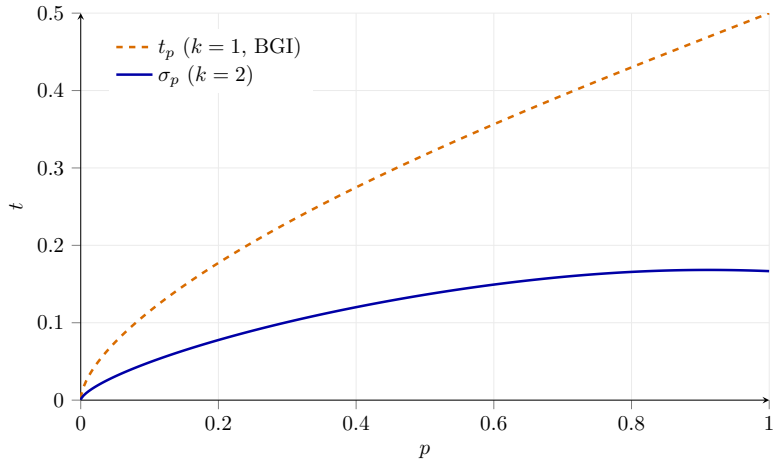


FIGURE 1. The points t_p from [2, Theorem 12] and σ_p from Theorem 2, computed numerically. For t below the curve, the extremals for $\Phi_1(p,t)$, respectively $\Phi_2(p,t)$, have one zero in $\mathbb{D}$; above it they have none.

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UNIVERSITY OF OSLO, OSLO, NORWAY
Email address: oistein.sovik@usit.uio.no